

Isotope Ratio Mass Spectrometer Data Analyzer

Data import Diagnostics **Calibration** Data Processing

Calibration

Standard Selection

Select Standards to Filter Data:

SHP2L ×



Outlier Detection

Set Sigma Level for standard's Outlier Exclusion

1.00



Set IQR Multiplier for standard's Outlier Exclusion

1.50



Choose Outlier Detection Method

Z-Score ▼

☒ Outliers independence (standards) ?

Visualization

Choose a parameter to color the dots:

d18O values ▼

Choose the Z-axis for 3D chart:

Signal Intensity ▼

Precision Date Range

Select date range for precision calculations (standards only):

2026/01/26 – 2026/03/10

Precision calculations use standards dated 2026-01-26 to 2026-03-10.

Calibrate results

☐ Apply linearity correction

☐ Use Samp-Ref intensity difference (Cycle 1, collector 44) ?

Precision and Averages for SHP2L (Excluding Outliers)

d13C Precision: 0.050‰

d18O Precision: 0.054‰

d13C Precision (linearity corrected): 0.040‰

d13C Average: -0.876‰

Line 1 precision: d13C 0.046‰ | d18O 0.038‰

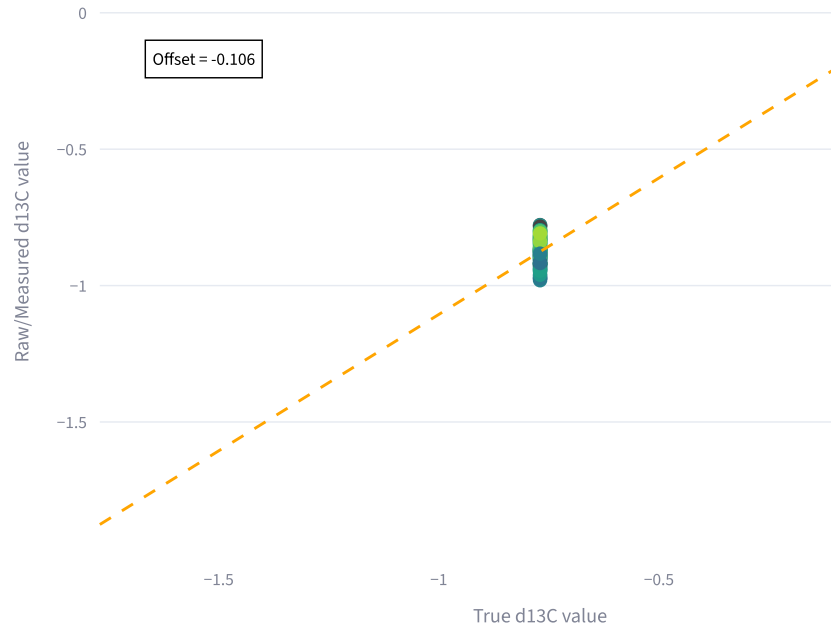
Line 2 precision: d13C 0.053‰ | d18O 0.063‰

Standards included: d13C 100 / 128 (78.1%) | d18O 106 / 128 (82.8%)

d18O Precision (linearity corrected): 0.048‰

d18O Average: -6.115‰

Single Anchor Calibration for VPDB(13C)



Single Anchor Calibration for VSMOW(18O)

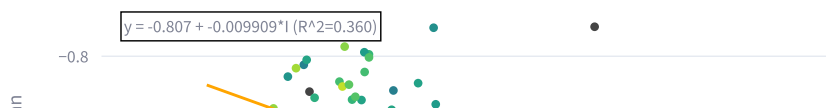


Calibration 3D Chart (Z-axis: Signal Intensity)

Linearity Correction

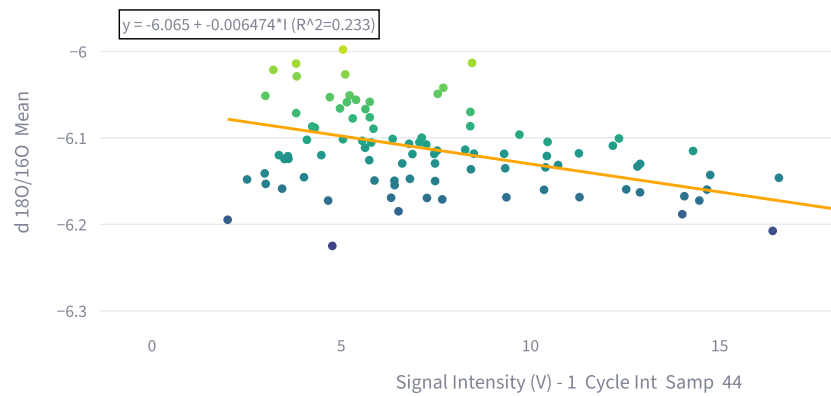
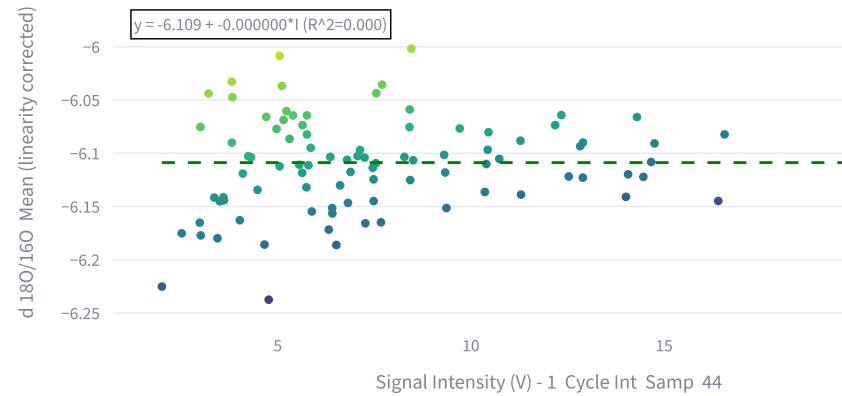
Linearity intensity basis: 1 Cycle Int Samp 44

Linearity (Standards): d 13C/12C Mean vs Intensity



Linearity (Standards): d 13C/12C Mean vs Intensity (Corrected)



Linearity (Standards): d 18O/16O Mean vs Intensity**Linearity (Standards): d 18O/16O Mean vs Intensity (Corrected)**

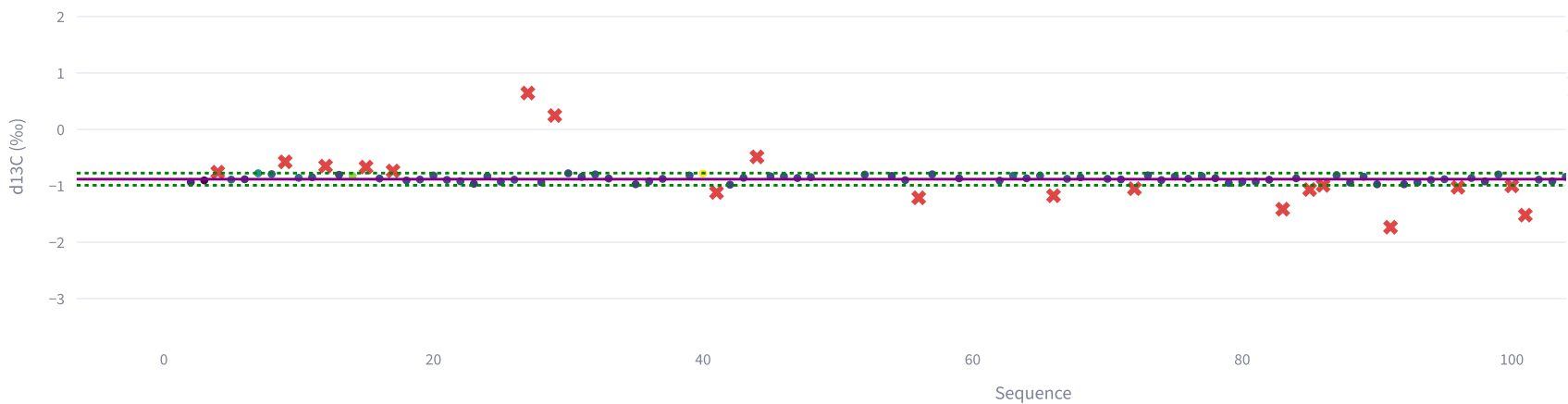
Identified Outliers for SHP2L

d13C Outliers:

d18O Outliers:

	d 13C/12C Mean		d 18O/16O Mean
15		10	
40		30	
51		40	
66		51	
76		61	

SHP2L d13C Calibration Values (Z-Score Method)



SHP2L d18O Calibration Values (Z-Score Method)

